EXPERIMENT NO: 5

Data Collection and Initial Exploration

PROGRAM:

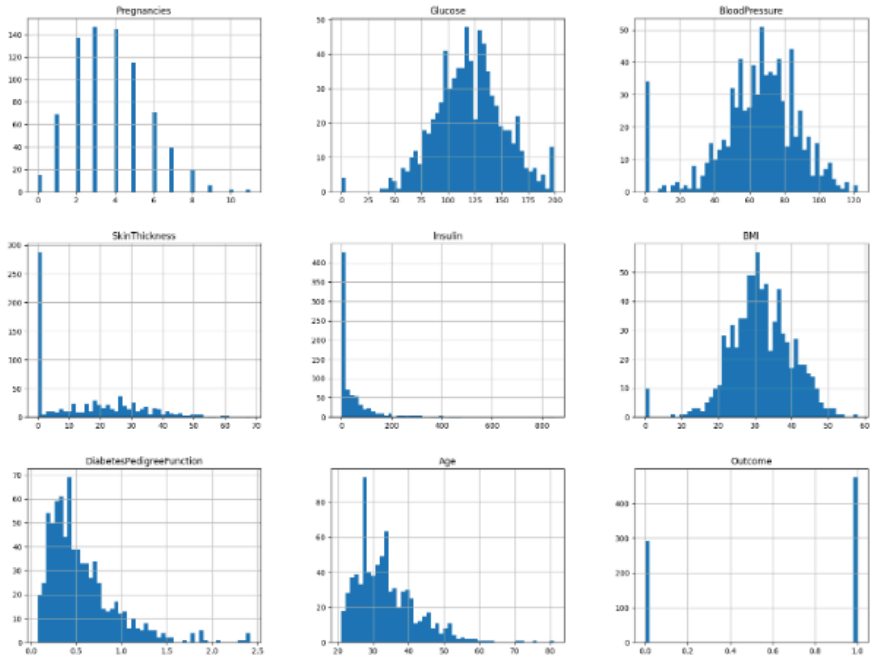
```
import pandas as pd
# Load the dataset
diabetes_df = pd.read_csv("diabetes.csv")
# Display the first few rows
print(diabetes_df.head())
```

OUTPUT:

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      3.768229  120.165365      64.972656      15.522135   44.391927
std       1.946060   32.387525      23.193434      15.576602   84.220647
min       0.000000    0.000000       0.000000       0.000000    0.000000
25%       2.000000   98.000000      53.000000       0.000000    0.000000
50%       4.000000  119.000000      67.000000      13.000000    5.000000
75%       5.000000  140.000000      80.000000      27.000000   55.250000
max      11.000000  199.000000     122.000000      68.000000  846.000000
```

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.616276	0.563865	33.558594	0.618490
std	8.581108	0.384463	8.546778	0.486074
min	0.000000	0.078000	21.000000	0.000000
25%	26.450000	0.299750	27.000000	0.000000
50%	31.300000	0.454500	32.000000	1.000000
75%	37.100000	0.717000	38.000000	1.000000
max	58.000000	2.420000	81.000000	1.000000


Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome

